

Acquisition

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Understanding Data Recovery Workstations and Software

- Investigations are conducted on a computer forensics lab (or data-recovery lab)
- Computer forensics and data-recovery are related but different
- **Computer forensics workstation**
 - Specially configured personal computer
 - Loaded with additional bays and forensics software
- To avoid altering the evidence use:
 - Forensics boot floppy disk
 - Write-blockers devices

Disk Interfaces

Following are the four types of interfaces that you can use to connect hard disks to motherboards.

- Small Computer System Interface (SCSI)
- Parallel Advanced Technology Attachment (PATA)
- Serial Advanced Technology Attachment (SATA)
- NVMe (Non-volatile Memory Express)

SCSI



PAT
A



SATA



NVM
e

Setting Up your Computer for Computer Forensics

- Basic requirements
 - A workstation running Windows
 - A write-blocker device
 - Computer forensics acquisition tool
 - Computer forensics analysis tool
 - Target drive to receive the source or suspect disk data
 - Spare PATA or SATA ports
 - USB ports

Setting Up your Computer for Computer Forensics (continued)

- Additional useful items
 - Network interface card (NIC)
 - Extra USB ports
 - FireWire 400/800 ports
 - SCSI card
 - Disk editor tool
 - Text editor tool
 - Graphics viewer program
 - Other specialized viewing tools

Conducting an Investigation

- Gather resources identified in investigation plan
- Items needed
 - Original storage media
 - Evidence custody form
 - Evidence container for the storage media
 - Bit-stream imaging tool
 - Forensic workstation to copy and examine your evidence
 - Securable evidence locker, cabinet, or safe

Gathering the Evidence

- Avoid damaging the evidence
- Steps
 - Meet the IT manager to interview him
 - Fill out the evidence form, have the IT manager sign
 - Place the evidence in a secure container
 - Complete the evidence custody form
 - Carry the evidence to the computer forensics lab
 - Create forensics copies (if possible)
 - Secure evidence by locking the container

Acquisition Process Overview

- Digital evidence acquisition: steps, tools, documentation.
- Maintains authenticity and reliability.

Role of Write-Blockers

- Prevents writing/modification of original data during acquisition.
- Ensures read-only access.

Types of Write-Blockers

- Hardware Write-Blockers: Tableau, WiebeTech.
- Software Write-Blockers: OS-level protection.

Importance of Write-Blockers

- Prevents contamination of evidence.
- Essential in legal proceedings and forensic standards.

Definition of Imaging

- Imaging: Creating an exact copy of digital storage.
- Preserves files, metadata, slack space, and deleted data.

Types of Imaging Techniques

1. Bit-by-bit imaging
2. Logical imaging
3. Live imaging

Bit-by-Bit Imaging

- Copies every sector including deleted/unallocated space.
- Highly reliable and court-accepted.

Logical Imaging

- Captures only visible files and folders.
- Used when full disk imaging isn't possible.

Live Imaging

- Used when the system is running and shutdown is not possible.
- Risk of volatile data loss.

Imaging Tools (FTK, dd, EnCase)

- Popular tools:
- FTK Imager, dd command, EnCase, Guymager, Clonezilla.

Evidence Integrity

- Evidence must remain unaltered.
- Integrity ensures admissibility in court.

Hashing Methods (MD5, SHA-1, SHA-256)

- Hash values are used for verification.
- Example: MD5, SHA-1, SHA-256.

Chain of Custody

- Documentation of who handled the evidence, when, and why.
- Critical for legal authenticity.

Standard Operating Procedure (SOP)

Overview

- Outlined procedures to ensure secure and legal evidence acquisition.

Pre-Acquisition Procedures

- Identify device, document system state, photograph setup.
- Prepare write-blockers and tools.

Acquisition Procedures

- Connect via write-blocker, capture image, calculate hash value.
- Save image securely.

Post-Acquisition Procedures

- Verify hash, label evidence, store securely.
- Update chain of custody records.

Preservation of Evidence

- Store evidence in tamper-proof bags.
- Maintain backups and access logs.

Challenges in Evidence Acquisition

- Encrypted drives, damaged disks, live systems, large storage capacities.

Best Practices & Legal Compliance

- Use certified tools, maintain logs, comply with ISO/IEC 27037 standards.

Analyzing Your Digital Evidence

- Your job is to recover data from:
 - Deleted files
 - File fragments
 - Complete files
- Deleted files linger on the disk until new data is saved on the same physical location

Introduction to Data Recovery and Carving

- Overview of data recovery and file carving.
- Used to retrieve deleted, damaged, or hidden data.
- Critical in digital forensic investigations.

Importance of Data Recovery in Forensics

- Helps uncover evidence from digital devices.
- Supports legal cases and investigations.
- Recovers deleted, intentionally hidden, or corrupted data.

What is Data Recovery?

- Process of retrieving lost or deleted data from storage devices.
- Uses file system structures like FAT, NTFS, ext.
- Often used when file paths and metadata are intact.

What is Data Carving?

- Technique to extract data fragments without using file system metadata.
- Identifies files using patterns like headers and footers.
- Useful in damaged or partially overwritten disks.

Difference: Data Recovery vs Data Carving

- Data Recovery: Uses file system metadata.
- Data Carving: Works without metadata.
- Recovery restores complete files; carving extracts fragments.

Types of Data Recovery

- Logical Recovery: Recovers data using OS tools.
- Physical Recovery: Involves deep disk analysis.
- Includes undelete, partition recovery, and disk imaging.

File Carving Methods

- File Signature-Based: Uses known file headers.
- Content-Based: Searches for patterns within raw data.
- Fragment Reassembly: Rebuilds fragmented files.

Header/Footer Carving

- Identifies files using known header and footer patterns.
- Example: JPEG files start with FF D8 and end with FF D9.
- Simple but may miss fragmented files.

File Structure Based Carving

- Uses internal file structure and format specifications.
- Example: Reconstructing file fragments using file type rules.
- More accurate but complex.

Challenges in Data Recovery & Carving

- Data fragmentation and corruption.
- Lack of metadata in carving.
- Risk of false positives and large data volume to analyze.